

the dangers of pregnancy; women on birth control pills have mortality rates about one-half those of pregnant women. Birth control pills also decrease the risk of ovarian and endometrial cancers. No hormonal contraceptives are available for men.

Sterilization is the permanent prevention of gamete production or release. For women, the most common method is **tubal ligation**, the sealing shut or tying off (ligating) of a section of each oviduct to prevent eggs from traveling into the uterus. Similarly, **vasectomy** in men is the cutting and tying off of each vas deferens to prevent sperm from entering the urethra. Sex hormone secretion and sexual function are unaffected by both procedures, with no change in menstrual cycles in females or ejaculate volume in males. Although tubal ligation and vasectomy are considered permanent, both procedures can in many cases be reversed by microsurgery.

The termination of a pregnancy in progress is called **abortion**. Spontaneous abortion, or *miscarriage*, is very common; it occurs in as many as one-third of all pregnancies, often before the woman is even aware she is pregnant. In addition, each year about 700,000 women in the United States choose to have an abortion performed by a physician.

A drug called mifepristone, or RU486, can terminate a pregnancy nonsurgically within the first 7 weeks. RU486 blocks progesterone receptors in the uterus, thus preventing progesterone from maintaining the pregnancy. It is taken with a small amount of prostaglandin to induce uterine contractions.

Modern Reproductive Technologies

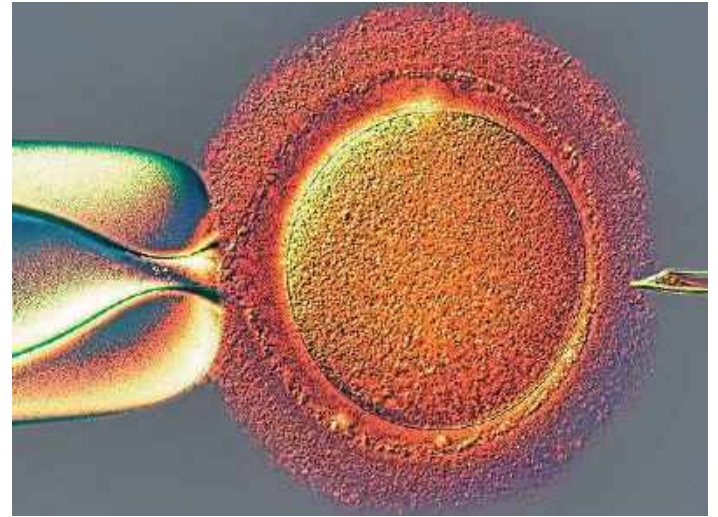
Recent scientific and technological advances have made it possible to address many reproductive problems, including genetic diseases and infertility.

Infertility and in Vitro Fertilization

Infertility—an inability to conceive offspring—is quite common, affecting about one in ten couples in the United States and worldwide. The likelihood of infertility is nearly the same for men and women and causes vary. For women, however, the risk of reproductive difficulties, as well as genetic abnormalities of the fetus, increases steadily past age 35. Evidence suggests that the prolonged period of time oocytes spend in meiosis is largely responsible for this increased risk.

Among preventable causes of infertility, sexually transmitted infections (STIs) are the most significant. In women 15–24 years old, approximately 830,000 cases of chlamydia and gonorrhea are reported annually in the United States. The actual number of women infected with the chlamydia or gonorrhea bacterium is considerably higher because most women with these infections have no symptoms and are therefore unaware of their infection.

▼ **Figure 46.21 In vitro fertilization (IVF).** In this form of IVF, a technician holds the egg in place with a pipette (left) and uses a very fine needle (right) to inject one sperm into the egg cytoplasm (colorized LM).



Up to 40% of women who remain untreated for either chlamydia or gonorrhea develop an inflammatory disorder that can scar the oviduct, substantially increasing the risk of a tubal, or ectopic, pregnancy. Rather than implanting in the uterus, the embryo becomes lodged in the oviduct (fallopian tube) where fertilization occurred. Such pregnancies cannot be sustained and may rupture the oviduct, resulting in serious internal bleeding.

Some forms of infertility are treatable. Hormone therapy can sometimes increase sperm or egg production, and surgery can often correct ducts that formed improperly or have become blocked. In some cases, doctors recommend **in vitro fertilization (IVF)**, which involves combining oocytes and sperm in the laboratory. Fertilized eggs are incubated until they form eight or more cells and are then transferred to the woman's uterus for implantation. If mature sperm are defective or low in number, a whole sperm or a spermatid nucleus may be injected directly into an oocyte (**Figure 46.21**). Though costly and not always successful, IVF procedures have enabled more than a million couples to conceive children.

Detecting Disorders During Pregnancy

Many developmental problems and genetic diseases can now be diagnosed while the fetus is in the uterus. Ultrasound imaging, which generates images using sound frequencies above the normal hearing range, is commonly used to analyze the fetus's size and condition. In amniocentesis and chorionic villus sampling, a needle is used to obtain fetal cells from either fluid or tissue, respectively, surrounding the embryo; these cells then provide the basis for genetic analysis (see Figure 14.19).

A new reproductive technology makes use of a pregnant mother's blood to analyze the genome of her fetus. As discussed in Concept 14.4, a pregnant woman's blood contains